Irtifa Haider

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RESEARCH INTEREST

Machine Learning; Natural Language Processing (NLP);Large Language Models (LLMs); Data Mining; Information Retrieval

EDUCATION

Master of Science (M.Sc.) in Computer Science and Engineering **Sept 2025 – Present**
Jahangirnagar University, Savar, Dhaka, Bangladesh

Bachelor of Science (B.Sc.) in Computer Science and Engineering **Mar 2020 – Sept 2025**
Jahangirnagar University, Savar, Dhaka, Bangladesh

- **CGPA:** 3.43/4.00
- Thesis: *The Star Paradigm: Redefining Social Network Clustering*.
- Relevant Coursework: Artificial Intelligence, Machine Learning, Pattern Recognition, Theory of Computation, Compiler Design, Database Systems, and Software Engineering.

Higher Secondary Certificate (HSC), Science **2019**
Government City College, Chattogram, Bangladesh
GPA: 4.83/5.00

Secondary School Certificate (SSC), Science **22017**
Agrabad Balika Bidyalay, Chattogram, Bangladesh
GPA: 5.00/5.00

PUBLICATIONS

Haider, I., Alam, S., Hossan, M. T., Anwar, M. M., & Aurpa, T. T. "A Comprehensive Text Optimization Approach to Bangla Summarization." *Proceedings of the Second Bangla Language Processing Workshop (BLP), co-located with the 14th International Joint Conference on Natural Language Processing (IJCNLP–AACL)*, 2025.

RESEARCH EXPERIENCE

Undergraduate Research **2024 – 2025**
Supervisor: Dr. Md. Musfique Anwar, Co-supervisor: Md. Tazel Hossan

- Conducted research on Bangla text summarization using transformer-based language models, including text optimization, dataset preprocessing, model training, hyperparameter tuning, and experimental evaluation.
- Research resulted in the publication: *"A Comprehensive Text Optimization Approach to Bangla Summarization"*, presented at BLP 2025 (IJCNLP–AACL 2025).

Undergraduate Thesis Research **2024 – 2025**
Supervisor: Dr. Sanjit Kumar Saha

- Conducted undergraduate thesis research on *"The Star Paradigm: Redefining Social Network Clustering"*, proposing and evaluating a graph-based clustering approach for social network analysis.

Research Work in Progress **2025 – Present**
Department of Computer Science and Engineering, Jahangirnagar University
Supervisor: Dr. Md. Ezharul Islam

- Developing an AI-enabled smart energy management and redistribution system for efficient power utilization.

ACADEMIC PROJECTS

Plant Classification | Python, scikit-learn, skimage, NumPy, pandas, matplotlib, joblib

- Developed a **fine-grained** plant species identification system for 18 morphologically similar species under heavy occlusion and illumination variance, achieving 92.3% accuracy (outperforming previous SVM-based baselines).
- Designed and implemented a complete ML pipeline from scratch: custom data augmentation, multi-scale **HOG + color histograms**, RBF-SVM with hierarchical classification, and hyperparameter optimization via **Bayesian optimization**.

LivSmart: Smart Living Community App | Java, XML, Firebase, Dokka, JUnit, Mockito, MVVM

- Developed an Android app streamlining residential operations (service requests, payments, events, alerts).

Hybrid Smart Energy System (IoT Project) | C++, Arduino IDE, ESP32, IoT Sensors, Blynk, Solar Panel

- Prioritized critical loads to design a cost-effective **IoT system optimizing** solar, grid, and battery power distribution.

Learn and Teach Programming | C++

- Developed a C++ platform offering tutorials, code samples, and lessons to enhance programming skills.

Criss Cross Game & Calculator | C

- Built a classic two-player tic-tac-toe game with an integrated calculator for basic computations.

IT Training Hub | HTML, CSS, JavaScript, PHP, MySQL, MVC

- Developed a web platform to manage IT training courses, trainers, and participants.
- Implemented a robust **MVC architecture** for scalability.

PROFESSIONAL EXPERIENCE

QA Functional Tester (Internship) | A1QA **April 2026 – June 2026**

- Worked with various testing types including Smoke Testing, Acceptance Testing, and Manual Testing.
- Designed and executed test cases, performed test planning, identified bugs, and reported defects using Jira with proper documentation.

Coding Instructor | CoderPals **Sept 2024 – April 2026**

- Taught Scratch and Python through engaging, game-based, and project-based lessons for children.
- Simplified programming concepts to foster computational thinking, creativity, and problem-solving skills.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, PHP

AI Frameworks & Models: TensorFlow, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, matplotlib, Word2Vec, BERT, BanglaBERT, mBERT, XLM-RoBERTa (XLM-R), mT5, T5, GPT-4, Gemini, Llama, Qwen2.5

Web Development: HTML, CSS, React, Bootstrap, Django, REST API, Amazon Web Services (AWS)

Development Tools: Git, GitHub, Android Studio, VS Code, NetBeans, Eclipse, Anaconda, Visual Paradigm, CI/CD, JUnit, Mockito, MATLAB, Slack, Trello, MySQL, MongoDB, Firebase, Blynk

Productivity & Collaboration Tools: Microsoft Office Suite (Word, Excel, Access, PowerPoint), Toggl, Overleaf, JIRA

CERTIFICATES

Certifications: freeCodeCamp certifications in Responsive Web Design and JavaScript Algorithms & Data Structures (2023–2024); passed the Fundamental IT Engineer (FE) Exam (Apr 2024).

Bootcamp: Cython Codecraft: Data Structures & Algorithm, Learn & Hack Web Development Bootcamp 2023–2024

HONORS & AWARDS

- **CP:** Solved 400+ problems on Codeforces (*Irtifa_CSE-29*), AtCoder (*Irtifa_CSE29*), and LeetCode (*haiderIr*).
- **International Conference:** Attended and presented paper at the *14th International Joint Conference on Natural Language Processing (IJCNLP) & 4th Asia-Pacific Chapter of the Association for Computational Linguistics*. **2025**
- **International Collegiate Programming Contest (ICPC):** Competed in the preliminary round. ***2021***
- **National Collegiate Programming Contest (NCPC):** Participated in the preliminary round. ***2021***
- **Merit Scholarships:** Received Government Board Merit Scholarships in both *JSC (2014)* and *SSC (2017)*.

VOLUNTEER EXPERIENCES

- The *Deep Ecology and Snake Conservation Foundation on ecological conservation initiatives.* **2023–Present**
- Event support staff at the *National Collegiate Programming Contest (NCPC), Jahangirnagar University* . **2024**

REFERENCES

Dr. Md. Musfique Anwar
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Dr. Sanjit Kumar Saha
Professor
Department of Computer Science and Engineering
Jahangirnagar University, Dhaka-1342, Bangladesh
Email: sanjit@juniv.edu